CAPSTONE PROJECT 1

Enterprise IT Operations & Service Management Agentic Platform

AI-Powered Multi-Agent Architecture using Azure AI Foundry

1. Executive Summary

Modern global enterprises process over **25,000+ IT incidents per month** across multiple channels including ServiceNow, Microsoft Teams, email systems such as Microsoft Outlook, and enterprise monitoring platforms.

Traditional IT Service Management (ITSM) models rely heavily on manual triage, siloed knowledge bases, and reactive workflows. This results in:

- SLA violations
- Increased operational costs
- Slow resolution cycles

- Inconsistent service quality
- High L1/L2 support workload

This capstone project proposes a **Multi-Agent AI-Powered Enterprise IT Operations Platform** built using **Azure AI Foundry and Azure-native services** to automate incident lifecycle management, enhance decision intelligence, and significantly improve SLA compliance.

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2. Problem Statement

A global enterprise receives over **25,000 IT incidents monthly** through multiple channels.

Current Operational Challenges

- Manual ticket triaging
- Slow incident resolution
- Knowledge silos across departments
- SLA violations
- High support costs
- Limited workflow automation
- Poor visibility into operational performance

The organization requires an **intelligent, scalable, AI-driven IT Operations Platform** capable of autonomous decision-making and orchestration.

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3. Proposed Solution: AI-Powered Agentic IT Operations Platform

Architecture Overview

The solution leverages a **Multi-Agent Architecture** built on:

- Azure AI Foundry
- Azure OpenAI Service (GPT-4o)
- Azure AI Search
- Azure Cosmos DB
- Azure Functions
- Azure Monitor
- Application Insights

- Azure Key Vault
- Power Automate

External Integrations

- ServiceNow
- Microsoft Teams
- SharePoint
- Microsoft Outlook
- CMDB APIs
- Active Directory

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4. Multi-Agent Architecture Design

The platform consists of specialized AI agents:

Agent	Responsibility
Intake Agent	Captures incidents from Teams, Email, ServiceNow
Classification Agent	Intent detection, priority tagging
Knowledge Agent	Searches SharePoint & AI Search indexes
CMDB Agent	Queries configuration & asset data
Automation Agent	Executes workflows via Power Automate
Monitoring Agent	Retrieves logs from Azure Monitor
Security Agent	Investigates suspicious activities
SLA Agent	Tracks SLA deadlines & escalations
Reporting Agent	Generates dashboards & executive summaries

Agents collaborate using orchestration logic powered by GPT-4o reasoning.

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5. System Workflow

1. Incident received via ServiceNow / Teams / Email

2. GPT-4o performs intent classification
3. Priority determined (P1–P4)
4. Knowledge base queried via Azure AI Search
5. CMDB validated for asset context
6. Automation workflow triggered (if applicable)
7. SLA monitored continuously
8. Escalation triggered for high severity
9. Dashboard updated in real time

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6. Business Use Cases (Enterprise Scenarios)

Case 1: VPN Not Working

Problem: Employee unable to connect Business Applications.

AI Solution Flow:

- Intent detection (Network Issue)
- KB search via Azure AI Search
- Auto-recommend troubleshooting steps
- Auto-create ticket in **ServiceNow**

Case 2: Password Reset

Problem: Locked account.

AI Solution:

- Identity verification
- Trigger reset via KB search workflow
- Notify user

Case 3: Server Down (P1 Incident)

Problem: Production outage.

Solution:

- Immediate classification as P1 & trigger case at **Service Now**
- Notify support
- If needed - Trigger outage war-room workflow

Case 4: Error – Code & Bug fixing

Solution:

- Check the Error
- Automatically Connect to respective tools
- Provide the solution & confirm with user.

Case 5: Network Latency

Solution:

- Query Azure Monitor logs
- Perform root cause analysis
- Suggest infrastructure scaling

Case 6: Printer Failure

Solution:

- Device diagnostics
- Generate **ServiceNow ticket**
- Assign field support & revert end user

Case 7: Security Alert

Problem: Suspicious login detected.

Solution:

- Security Agent investigates logs
- Cross-check identity anomalies
- Escalate to SOC team

Case 8: SharePoint-Documents/Files management

Solution:

- Document Retrieval & analysis
- Document Upload & Versioning services
- Broadcast status notification

Case 9: Database Performance Issue

Solution:

- Analyze SQL logs
- Detect slow queries
- Recommend indexing or scaling

Case 10: Reasoning agent with ReAct

Solution:

- Agent can connect with Enterprise Application
- Auto-generate user responses & Queries
- Also, it can generate Structured response
- And Agent can Switch to IT/HR/ServiceNow related queries automatically.

Case 11: Agent can connect to enterprise RAG

Solution:

- Agent can connect with external APIs (Weather API+ Live News+ Any Opensource Communities)
- Notify users
- Agent should handle the semantic + hybrid search capabilities

Case 12: Weekly Incident Summary

Solution:

- AI-generated operational summary

- Visual dashboard creation
- provide report to users/management/leadership

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8. Performance & Business Impact

Metric	Improvement
L1 Workload	↓ 70%
Incident Resolution Time	↓ 50%
SLA Compliance	↑ Significant Improvement
Operational Cost	↓ 30–40%
Automation Coverage	↑ 65%

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9. Real-Time Integrations

- ServiceNow
- Microsoft Teams
- SharePoint
- Outlook
- Azure Monitor
- CMDB
- Active Directory

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10. Strategic Value to Enterprise

This platform transforms IT operations from:

Reactive Support → Intelligent Autonomous Operations

It establishes:

- AI-driven decision intelligence
- Cross-system orchestration
- Predictive SLA management

- Executive-level operational visibility
- Scalable multi-agent automation

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11. Conclusion

The **Enterprise IT Operations & Service Management Agentic Platform** demonstrates how Azure AI Foundry and Multi-Agent Architecture can revolutionize enterprise IT.

By integrating intelligent automation, contextual reasoning, and real-time orchestration, the organization achieves:

- 70% reduction in L1 workload
- 50% faster resolution
- Reduced SLA violations
- Lower operational costs
- Enhanced user satisfaction

This solution positions the enterprise toward **Autonomous IT Operations (AIOps 2.0)** and future-ready digital transformation.

Important Submission Instructions:

Deliverable Format

Participants must prepare and submit their capstone project report in either of the following formats:

- Microsoft Word Document (.docx)
- PDF Document (.pdf)

Naming Convention

Please save your file using the following naming format:

YOURNAME_EMPID.docx

OR

YOURNAME_EMPID.pdf

Example:

AjayKumar_123456.docx

or

AjayKumar_123456.pdf

Submission Guidelines

1. Ensure all sections of the capstone project are completed.
2. Include architecture diagrams, screenshots, implementation details, case solutions, and conclusions.
3. Verify that the document follows the prescribed naming convention.
4. Upload the final document to the LMS Portal folder provided by the trainer.
5. Submit before the specified deadline.

Mandatory Sections

- Cover Page
- Problem Statement
- Project Objectives
- Solution Architecture
- Tools & Technologies Used
- Implementation Details
- All Business Cases and Solutions – with Screenshots
- Security & Governance Implementation
- Monitoring & Observability Dashboard
- Performance Optimization Strategy
- Real-Time Integrations
- Conclusion
- References (if applicable)

Evaluation Criteria

- Architecture Design
- Agent Framework Implementation
- RAG Implementation
- Multi-Agent Orchestration
- Enterprise Integrations
- Security & Governance

- Monitoring & Observability
- Innovation & Business Value
- Documentation Quality
- Final Demonstration

Failure to follow the naming convention or submission guidelines may impact the evaluation process.